

HUNDRED DAYS OF CODE

DAY 15

QUESTION : Write a menu-driven C program using switch-case to input a positive integer and Check whether all digits are even.

- **E.g. 1:** Number: 12345, Output: All digits not even
- **E.g. 2:** Number: 24680, Output: All digits are even give me the input, output, algorithm, pseudocode, flowchart, test case table , required c program for this assignment.

INPUT: A positive integer entered by the user

OUTPUT: Display whether all the digits of the given number are even.

ALGORITHM: 1. Start.

2. Input a positive integer num.

3. Set $n = \text{num}$ and $\text{allEven} = 1$.

4. Repeat until n becomes 0:

- Find the last digit using $\text{digit} = n \% 10$.
- Use a switch-case on $\text{digit} \% 2$.
- If the result is 1, the digit is odd, so set $\text{allEven} = 0$.
- Remove the last digit using $n = n / 10$.

5. If $\text{allEven} == 1$, display **“All digits are even”**.

6. Otherwise, display **“All digits not even”**.

7. Stop.

PSEUDOCODE:

START

INPUT num

$n \leftarrow \text{num}$

$\text{allEven} \leftarrow 1$

WHILE $n > 0$

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digit $\leftarrow$ n MOD 10

SWITCH (digit MOD 2)

CASE 0:

Continue checking

BREAK

CASE 1:

allEven $\leftarrow$ 0

BREAK

END SWITCH

n $\leftarrow$ n / 10

END WHILE

IF allEven = 1 THEN

PRINT "All digits are even"

ELSE

PRINT "All digits not even"

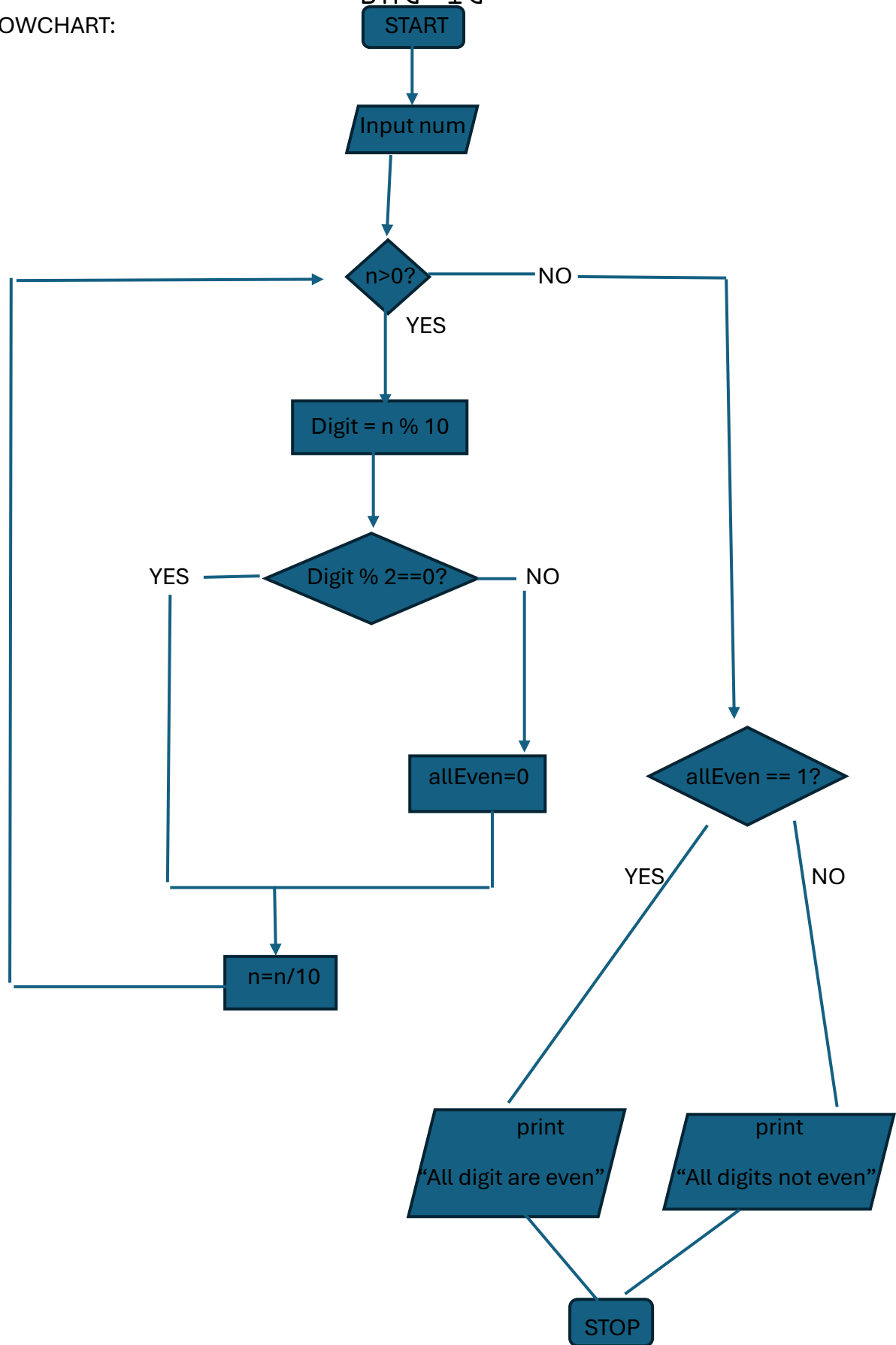
END IF

STOP

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FLOWCHART:



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TEST CASE TABLE:

TEST CASE	INPUT NUMBER	DIGITS CHECKED	EXPECTED OUTPUT	RESULT
1	12345	1,2,3,4,5	ALL DIGITS ARE NOT EVEN	PASS
2	24680	2,4,6,8,0	ALL DIGITS ARE EVEN	PASS
3	2222	2,2,2,2	ALL DIGITS ARE EVEN	PASS

REQUIRED C PROGRAM:

```
#include <stdio.h>
```

```
int main()
```

```
{
```

```
    int num, n, digit, allEven = 1;
```

```
    printf("Enter a positive integer: ");
```

```
    scanf("%d", &num);
```

```
    n = num;
```

```
    while (n > 0)
```

```
    {
```

```
        digit = n % 10;
```

```
        switch (digit % 2)
```

```
        {
```

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```
case 0:

    /* Digit is even */

    break;


case 1:

    /* Digit is odd */

    allEven = 0;

    break;

}


n = n / 10;

}


if (allEven == 1)

    printf("All digits are even\n");

else

    printf("All digits not even\n");


return 0;

}
```

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COMPILATION AND EXECUTION:

```
(kali㉿kali)-[~]  
$ vi positive.c  
  
(kali㉿kali)-[~]  
$ gcc positive.c -o positive  
  
(kali㉿kali)-[~]  
$ ./positive  
Enter a positive integer: 12345  
All digits not even  
  
(kali㉿kali)-[~]  
$ ./positive  
Enter a positive integer: 24680  
All digits are even  
  
(kali㉿kali)-[~]  
$ ./positive  
Enter a positive integer: 2222  
All digits are even  
  
(kali㉿kali)-[~]  
$
```
